

scaled model, emulating the combination of small-scale fine-tuning and large-scale pre-trained knowledge; d) the EFT *down-scaled* model, emulating the combination of large-scale fine-tuning with small-scale pre-trained knowledge. For example, for the Llama-2 experiments, we sample from a) Llama-2-chat 7B; b) Llama-2-chat 70B; c) up-scaled EFT with Llama-2-base 70B as the pre-trained model and Llama-2-chat 7B/Llama-2-base 7B as the implicit reward; and d) down-scaled EFT with Llama-2-base 7B as the pre-trained model and Llama-2-chat 70B/Llama-2-base 70B as the implicit reward. All experiments use temperature sampling with temperature 1.0, without top-p or top-k (except when specified otherwise).

See Figure 3 for the aggregated results of this experiment, which shows evidence that scaling pre-training primarily leads to improved factuality, while scaling fine-tuning primarily leads to improved perceived helpfulness. See Figures 4 and 6 for the per-model and per-dataset results. Results are normalized against the performance of the small and large fine-tuned models alone (which are essentially lower and upper bounds on performance); a value of 0.0 corresponds to small fine-tuned model performance, while a value of 1.0 corresponds to large fine-tuned model performance. Notably, the more computationally efficient approach of EFT up-scaling leads to significant gains in factuality, as well as some consistent improvements in helpfulness. Section 4.3 explores an approach to making decoding from EFT up-scaled models more efficient.

4.2 EFT Enables Dynamic Test-Time Reward Interpolation

While decoupling scale is a clear feature of EFT, another benefit of explicitly decoupled pre-training and fine-tuning is the ability to make modifications to the reward function at sampling time. Consider the case of competing fine-tuning objectives, such as the objectives of helpfulness and harmlessness (Bai et al., 2022); some user queries (‘How can I steal my neighbor’s guitars?’), providing an answer that helps the user with their goal is directly at odds with providing a harmless (or safe) answer. Thus, one view of fine-tuning general dialogue agents is attempting to provide maximum helpfulness at a particular budget of harmfulness. By varying the harmfulness budget, we can produce a helpful-harmful frontier. However, existing fine-tuning procedures *bake in* the particular desired tradeoff between helpfulness and harmfulness at fine-tuning time, and this tradeoff cannot be easily modified at sampling time.

In contrast, with emulated fine-tuning, such test-time modulation of the reward is natural and straightforward. Figure 5 shows the results of interpolating between helpfulness and harmlessness at 7B pre-training and fine-tuning scale, as well as with up-scaling the pre-trained model to 70B. We see clear, smooth frontiers, and up-scaling provides a Pareto improvement, **all without retraining to each tradeoff**.

To interpolate behaviors at test time with EFT, we assume that two small-scale fine-tuned models exist, one fine-tuned for pure helpfulness π_{help} , one for pure harmlessness π_{safe} . For this experiment, we fine-tune these two models with DPO using Llama-2-7B as the base model, and the helpful-base and harmless-base splits of the Anthropic-HH dataset (Bai et al., 2022). At test time, instead of using a single reward function $r_{\pi}^M(x, y)$ in Equation 4, we use the interpolated reward $r_{\lambda}^M(x, y) = \lambda r_{\text{help}}^M(x, y) + (1 - \lambda) \pi_{\text{safe}}^M$, where $\lambda = 1$ corresponds to pure helpfulness, and $\lambda = 0$ pure harmlessness. Sampling with $\lambda \in (0, 1)$ corresponds to some mixture of helpful and harmless. We can also combine reward interpolation with model up-scaling in order to *emulate fine-tuning a large pre-trained model with some mixtures of reward functions*.

4.3 Efficient Sampling from Up-scaled Models with Speculative Decoding

Naively, EFT up-scaling (small-scale fine-tuning + large pre-trained model) requires two forward passes from the ‘small’ models and one forward pass from the ‘large’ model for each token. Yet the size asymmetry

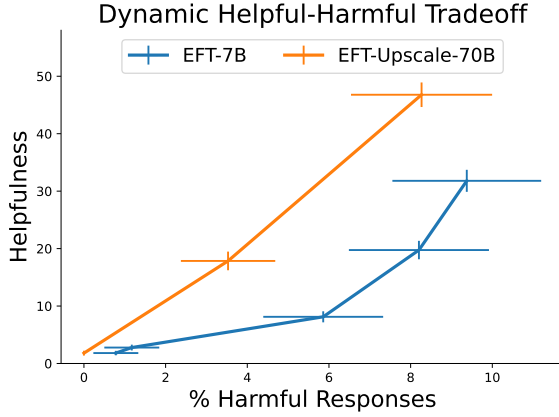


Figure 5: **Dynamically adjusting the desired tradeoff between helpfulness and harmlessness without retraining.** We use EFT to interpolate between two implicit rewards for helpfulness and harmlessness and plot GPT-4-evaluated helpfulness and fraction of responses that are harmful on Anthropic-HH prompts. Combining reward interpolation with up-scaling enables a Pareto improvement in the frontier, **all without fine-tuning**. Error bars are one standard error.

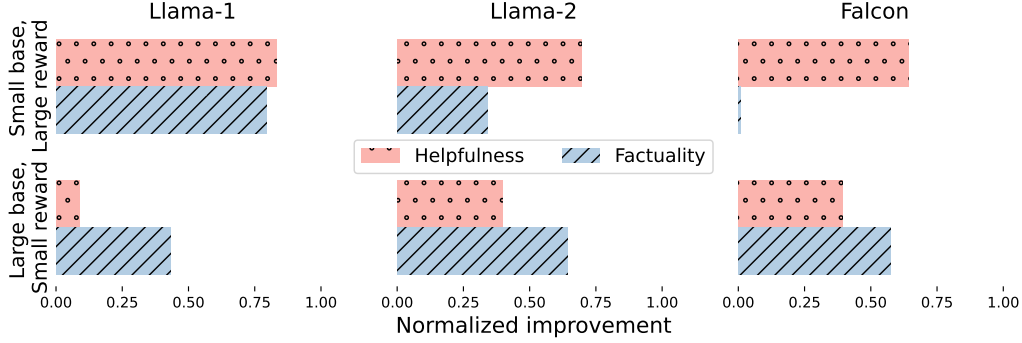


Figure 6: Normalized improvements in **factuality** and **helpfulness** from emulated fine-tuning on prompts from ELI5 dataset. Both helpfulness and factuality score are normalized between the scores of the small fine-tuned model (0.0) and the large fine-tuned model (1.0). Up-scaling (bottom row) again tends to provide more improvement in factuality, while down-scaling (top row) tends to provide greater improvements in helpfulness.

of EFT makes speculative decoding (Chen et al., 2023a) a natural choice to accelerate inference. Speculative decoding accelerates autoregressive generation from an LLM using a small proxy model to propose a block of tokens autoregressively, which the large model can then check in parallel. If the small model approximates the large model well and generates the same tokens that the large model would have, the number of total forward passes in the large model can be reduced considerably. For EFT up-scaling, we hypothesize that the small fine-tuned model alone might approximate the up-scaled model for most tokens; we verify this hypothesis qualitatively in Figure 7, which shows that the total variation distance between the small fine-tuned model and the up-scaled model is small for most tokens, and very large for a few tokens. Thus, speculative decoding is likely to accelerate EFT up-scaling.

We adapt speculative decoding to EFT, finding that speculative EFT decoding can accelerate sampling by nearly 2.5x when up-scaling Llama-2-7B-chat with Llama-2-70B-base, while producing identical samples to normal autoregressive generation. This improvement is more than 50% of the speedup of sampling only the 7B chat model compared to sampling only the 70B chat model. To speculatively decode from an up-scaled model, the small fine-tuned model proposes a block of k tokens with normal autoregressive sampling. Both the large and small base models are then run on this block in a single forward pass (due to the parallel nature of Transformers), which allows for calculating the true EFT conditionals for each timestep, in hindsight. If sampling from the true conditionals produces the same tokens⁴, we simply continue and sample a new proposed block. In the case of a disagreement, we rewind generation back to the last token where the small fine-tuned model and complete up-scaled model agreed. If no tokens agree, we use the token sampled from the first true hindsight up-scaled conditional.

4.4 Conservative Decoding Strategies for Up-Scaled Models

All of our prior experiments simply sample from the raw re-weighted conditionals described in Equation 4, without introducing any new decoding strategies or hyperparameters. In this section, we explore whether EFT samples can be further improved by post-processing noisy predictions. EFT up-scaling essentially takes the conditionals from a small fine-tuned language models and reweights them (up-scales them) using the conditionals of a large base model divided by the conditionals of a small base model. However, the up-scaling ratio $\frac{p_{\text{base-large}}(x_t|x_{<t})}{p_{\text{base-small}}(x_t|x_{<t})}$ may become extremely large for low-probability (and possibly poorly-modeled) tokens, leading to problematically high probability assigned to low-quality tokens.

Truncation	None	0.95	0.9	0.8
Errors/prompt	0.300	0.289	0.352	0.348
Helpfulness	66.8	67.0	67.2	67.0

Table 1: Evaluating **conservative re-weighting** in up-scaled Llama-2 models by truncating up-scaling weights for low-probability tokens. Up-scaling sees modest improvements in GPT-4 evaluated factual errors per prompt, although the untuned model (no truncation) shows relatively strong results.

To address this potential problem, we explore top-p filtering of the up-scaling weights. See Table 1 for complete results, showing that top-p filtering of up-scaling weights produces mild improvements in factuality

⁴We set the random seed to be equal to the timestep, to ensure high-entropy conditionals are not penalized.

Hello! I'm happy to help you with your question. A cup of chopped cauliflower contains approximately 25-27[↑7 ↓9] calories. However, please note that the exact number of calories can vary depending on the size and weight[↑weight ↓fresh] of the cauliflower, as well as any seasonings or cooking methods used. Is there anything else I can help you with?

Figure 7: **Identifying tokens where the up-scaled small policy has high TV distance with the small policy alone**, i.e., significant probability mass is moved. Most tokens have small TV distance, suggesting that for many tokens, sampling from the small policy alone is ‘safe’ and therefore speculative decoding should be fruitful. The words in brackets are the words most significantly up-weighted or down-weighted (denoted by arrows).

Spec. Block size	None	2	4	8	16	70B policy	7B policy
Toks/sec (HH)	6.0	9.2	12.5	13.8	12.1	9.3	28.0
Toks/sec (ELI5)	6.1	9.5	13.2	15.1	14.2		

Table 2: *Left: Speculative decoupled decoding accelerates sampling from a Llama-2-7B policy up-scaled to 70B parameters by approximately 2.5 times.* Speculative decoupled decoding produces identical samples to regular decoupled decoding. Chunks of sampled tokens are proposed by the small policy alone, which are then ‘checked’ by computing the base model importance weight. *Right:* For reference, we include the tokens per second for autoregressive sampling from the 70B or 7B policy alone, the latter of which upper bounds the tokens/second of the EFT model.

and helpfulness compared to sampling from the unfiltered conditionals. To perform top-p filtering, we first compute the ‘top-p’ set of tokens from the conditional of only the small fine-tuned model, that is, the smallest set of tokens whose probability sums to over p . However, unlike conventional top-p decoding (Holtzman et al., 2020), we do not set the conditionals to other tokens to zero. Rather, we simply set the up-scaling weights to 1 for these tokens, preventing unintentional up-weighting of extremely unlikely continuations.

4.5 Comparing GPT-4 Factuality Judgments with Human Evaluators

While the usage of large language models for evaluating human preferences or helpfulness has been validated in several cases (Zheng et al., 2023; Dubois et al., 2023; Gilardi et al., 2023; Rafailov et al., 2023), their effectiveness at performing fact-checking for everyday topics has not been extensively studied. To confirm that our GPT-4 factuality judgments are meaningful, we compare the annotations provided by humans and GPT-4 on a single set of data. Details of the human label collection are provided in the Appendix. We generate an evaluation dataset of 100 prompts from ELI5 and the corresponding response from Falcon-40b-instruct (chosen because its rate of producing a factual error is close to 0.5, according to GPT-4). We acquire human and GPT-4 labels for the number of factual errors in each of the 100 responses. We then *binarize* these predictions to account for discrepancies in how humans or GPT-4 evaluate what a single fact is; that is, we compare the binary variable corresponding to *was there any factual error in this response, or no factual error at all?* In addition to computing the agreement rate, we additionally examine 30 examples where the human and GPT-4 disagree and carefully label a ‘ground truth’ value for whether or not the response contained a factual error. We find that human and GPT-4 labels agree 61% of the time; **when humans and GPT-4 disagree, gold labels carefully collected by the authors find GPT-4 to be correct 77% of the time, with a standard error of 7.8%**. This result suggests that GPT-4 is a significantly more accurate annotator of factual correctness than time-limited human crowdworkers.

5 Conclusion

Scaling up the two-stage pipeline of pre-training and fine-tuning (or ‘alignment’) continues to be the dominant strategy for building more powerful language systems. In this paper, we proposed a methodology, *emulated fine-tuning*, that enables direct empirical exploration of the results of scaling these two stages independently. Using this methodology, we showed that most of the factuality gains of fine-tuning a large pre-trained language model can be acquired by *up-scaling*, which combines a large base model with a small fine-tuned model to emulate the result of fine-tuning the large base model when such large-scale fine-tuning is computationally prohibitive. Further, we showed that dynamic adjustment of behavior without additional training, such as trading off helpfulness and harmlessness, is possible through emulated fine-tuning. Future work may use emulated fine-tuning to study additional dimensions of model capabilities to those in our experiments, interpolate between other test-time model behaviors without requiring additional tuning, or explore alternative methods for sampling from EFT-structured models to improve efficiency or performance.